

How disrupted spatial fluxes reshape food web structure along the river–sea continuum

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Abstract

Anthropic disturbances and ecosystem processes transcend the artificial boundaries between land, freshwater, and sea. While these systems are interconnected through species and material fluxes, they are often studied in isolation. In response, the concept of metaecosystem has emerged as a powerful framework to understand how fluxes among ecosystems shape their functions and dynamics. Disrupting these fluxes can have dramatic consequences that ripple across entire landscapes. In aquatic systems, dams and nutrient pollution are known to strongly affect fish communities. Yet, little is still known about how these effects propagate and potentially accumulate, through hydrographic basins and ultimately to the sea. Using bayesian spatio-temporal statistical models (R-INLA), we aim to analyse existing data on food-web structure across France, combined with environmental variables (dams, nutrients, ...), to quantify how breaking or increasing fluxes among systems disturbs food-web structure locally and how these disturbances spread throughout the metaecosystem.

Keywords: *foodweb ecology, stream network, spatial autocorrelation, bayesian statistics.*

Introduction

This is some text.

Multiple stressors

- Synergistic and antagonistic effects of multiple stressors (Jackson, Pawar, and Woodward 2021; Orr et al. 2024).
- Temperature can interact with nutrients and impact food web structure in a nonlinear way Binzer et al. (2012).
- (Danet et al. 2021).

Connectivity

- Small dispersal stabilizes through spatial structure and asynchrony (LeCraw, Kratina, and Srivastava 2014; McCann, Rasmussen, and Umbanhowar 2005; Pillai, Gonzalez, and Loreau 2011).
- Prey switching of mobile top predators can stabilize connected food webs through ‘spatial coupling’ (LeCraw, Kratina, and Srivastava 2014; McCann, Rasmussen, and Umbanhowar 2005; Pillai, Gonzalez, and Loreau 2011).
- At high dispersal/connectivity the stabilizing effect wanes as strongly connected patches behave as a single patch (LeCraw, Kratina, and Srivastava 2014; McCann, Rasmussen, and Umbanhowar 2005).
- Studies between pairs of the following connectivity, complexity and stability are common. However, studies investigating how these are interconnected are rare (LeCraw, Kratina, and Srivastava 2014).
- Connectivity has 4 dimensions: longitudinal, lateral, vertical and temporal (Grill et al. 2019).
- We identified five pressure factors to represent the main human interferences within the four dimensions of fluvial or river connectivity (Grill et al. 2019):
 - (1) river fragmentation (longitudinal);
 - (2) flow regulation (lateral and temporal);
 - (3) sediment trapping (longitudinal, lateral and vertical);
 - (4) water consumption (lateral, vertical and temporal);
 - (5) infrastructure development in riparian areas and floodplains (lateral and longitudinal).

Dams

- Decrease river connectivity / reduce migration (Dean et al. 2023; He et al. 2024; Pedersen et al. 2012).

- Reservoir, lotic from lentic conditions, can result in hypoxic conditions. Favor phytoplanktons and macrophytes (Ding et al. 2026; Dean et al. 2023).
- Reservoir increase residence time of nutrients upstream and decrease the downstream output, which can result in a decrease in primary productivity.
- Hydropeaking - that is, the sudden release of large water to answer the higher electric demand - can harm riverine community due to sudden change in water level and water flow.
- Cumulative impact of dams can reduce the nutrient input in deltas (Dean et al. 2023).

Methods

Food web inference

Infer missing fish lengths

Most length of fishes are measured but not all. Yet, we need individual fish length to determine their diet based on their ontogeny. We infer missing fish size using information on the given fishing operation. When there are more than three observations of the fish species, we use a truncated normal distribution (package `truncdist`; Nadarajah and Kotz (2006)). The mean and variance are given by the modes of the observed distribution of the fishing operation. The minimum and maximum of the distribution are given by the minimal and maximal length observed on the entire data set for that species. We do so to avoid inferring unrealistic lengths.

Get fish diet

The metaweb

Subsampling the metaweb to get local food webs

Results

Discussion

Appendix

Model DAG

Put model DAG here using targets.

Binzer, Amrei, Christian Guill, Ulrich Brose, and Björn C. Rall. 2012. “The Dynamics of Food Chains Under Climate Change and Nutrient Enrichment.”

- Philosophical Transactions of the Royal Society B: Biological Sciences* 367 (1605): 2935–44. <https://doi.org/10.1098/rstb.2012.0230>.
- Bonnaffé, Willem, Alain Danet, Camille Leclerc, Victor Frossard, Eric Edeline, and Arnaud Sentis. 2024. “The Interaction Between Warming and Enrichment Accelerates Food-Web Simplification in Freshwater Systems.” *Ecology Letters* 27 (8): e14480. <https://doi.org/10.1111/ele.14480>.
- Danet, Alain, Maud Mouchet, Willem Bonnaffé, Elisa Thébault, and Colin Fontaine. 2021. “Species Richness and Food-Web Structure Jointly Drive Community Biomass and Its Temporal Stability in Fish Communities.” *Ecology Letters* 24 (11): 2364–77. <https://doi.org/10.1111/ele.13857>.
- Dean, E. M., Dana M. Infante, Hao Yu, Arthur Cooper, Lizhu Wang, and Jared Ross. 2023. “Cumulative Effects of Dams on Migratory Fishes Across the Conterminous United States: Regional Patterns in Fish Responses to River Network Fragmentation.” *River Research and Applications* 39 (9): 1736–48. <https://doi.org/10.1002/rra.4173>.
- Ding, Na, Zunyi Xie, Xiaoming Wang, Zongming Wang, Miao Li, Beiming Cai, Qinghe Zhao, et al. 2026. “Freshwater Biodiversity Impacts of Global Hydropower Dams.” *Communications Earth & Environment*, February. <https://doi.org/10.1038/s43247-026-03263-y>.
- Grill, G., B. Lehner, M. Thieme, B. Geenen, D. Tickner, F. Antonelli, S. Babu, et al. 2019. “Mapping the World’s Free-Flowing Rivers.” *Nature* 569 (7755): 215–21. <https://doi.org/10.1038/s41586-019-1111-9>.
- He, Fengzhi, Christiane Zarfl, Klement Tockner, Julian D. Olden, Zilca Campos, Fábio Muniz, Jens-Christian Svenning, and Sonja C. Jähnig. 2024. “Hydropower Impacts on Riverine Biodiversity.” *Nature Reviews Earth & Environment* 5 (11): 755–72. <https://doi.org/10.1038/s43017-024-00596-0>.
- Jackson, Michelle C., Samraat Pawar, and Guy Woodward. 2021. “The Temporal Dynamics of Multiple Stressor Effects: From Individuals to Ecosystems.” *Trends in Ecology & Evolution* 36 (5): 402–10. <https://doi.org/10.1016/j.tree.2021.01.005>.
- LeCraw, Robin M., Pavel Kratina, and Diane S. Srivastava. 2014. “Food Web Complexity and Stability Across Habitat Connectivity Gradients.” *Oecologia* 176 (4): 903–15. <https://doi.org/10.1007/s00442-014-3083-7>.
- McCann, K. S., J. B. Rasmussen, and J. Umbanhowar. 2005. “The Dynamics of Spatially Coupled Food Webs.” *Ecology Letters* 8 (5): 513–23. <https://doi.org/10.1111/j.1461-0248.2005.00742.x>.
- Nadarajah, Saralees, and Samuel Kotz. 2006. “R Programs for Truncated Distributions.” *Journal of Statistical Software* 16 (August): 1–8. <https://doi.org/10.18637/jss.v016.c02>.
- Orr, James A., Samuel J. Macaulay, Adriana Mordente, Benjamin Burgess, Dania Albin, Julia G. Hunn, Katherin Restrepo-Sulez, et al. 2024. “Studying Interactions Among Anthropogenic Stressors in Freshwater Ecosystems: A Systematic Review of 2396 Multiple-Stressor Experiments.” *Ecology Letters* 27 (6): e14463. <https://doi.org/10.1111/ele.14463>.
- Pedersen, M. I., N. Jepsen, K. Aarestrup, A. Koed, S. Pedersen, and F. Økland. 2012. “Loss of European Silver Eel Passing a Hydropower Station.” *Journal*

of Applied Ichthyology 28 (2): 189–93. <https://doi.org/10.1111/j.1439-0426.2011.01913.x>.

Pillai, Pradeep, Andrew Gonzalez, and Michel Loreau. 2011. “Metacommunity Theory Explains the Emergence of Food Web Complexity.” *Proceedings of the National Academy of Sciences* 108 (48): 19293–98. <https://doi.org/10.1073/pnas.1106235108>.